

ARCHAEOLOGY

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages and pick up an archaeologist's trowel. In the seventh-century royal ship burial at Sutton Hoo, the remains of a lyre were found – the instrument of Anglo-Saxon bards. In tenth-century Denmark the Vikings built Trelleborg-type fortresses: their plan is a circle divided into four by two roads. Inscriptions on rune stones were carved in the Younger Futhark. Viking boats were built in the clinker technique, in which the hull planks overlap like roof tiles. Hedeby, or Haithabu, was a Viking trading center in the ninth and tenth centuries.

Dendrochronology dates the felling of timber by the pattern of annual growth rings. Graves laid out on an east–west line are most often linked by archaeologists with Christian burial rites, and the memento mori motif on tombstones reminds us of transience and the inevitability of death. DNA studies of Black Death victims confirmed that the plague was caused by the bacterium *Yersinia pestis*. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The remains of which instrument were found in the royal ship burial at Sutton Hoo (England, 7th century)?
 - a) A lute
 - b) An organ
 - c) Bagpipes
 - d) A lyre
2. 👁 What layout do the 10th-century Viking ring fortresses of the Trelleborg type in Denmark have?
 - a) A five-pointed star
 - b) A long rectangle along the shore
 - c) A circle divided into four by two roads
 - d) An irregular polygon fitted to the terrain
3. 📖 What script was used for inscriptions on Scandinavian runestones of the Viking Age?
 - a) The Younger Futhark
 - b) The Latin alphabet
 - c) Cyrillic
 - d) Cuneiform
4. 📅 Dendrochronology can determine the year excavated timber was felled based on...
 - a) the smell of the resin
 - b) the iron content of the wood
 - c) the pattern of annual growth rings
 - d) the color of the bark
5. 🚢 Viking boats were built using the clinker technique, meaning that...
 - a) the planks are joined edge to edge
 - b) the hull planks overlap like roof tiles
 - c) the hull is hollowed from a single trunk
 - d) the hull is covered with hide
6. 🏰 What was Hedeby (Haithabu) in the 9th–10th centuries?
 - a) A royal salt mine
 - b) A Benedictine monastery
 - c) A Viking trading center
 - d) A Teutonic castle on the Vistula
7. 💜 The "memento mori" motif on medieval tombs and in art is a reminder of...
 - a) good harvests and bountiful years
 - b) past military victories and glory
 - c) transience and the inevitability of death
 - d) the ancient lineage of a family
8. 🍀 DNA studies of Black Death victims (e.g., from the East Smithfield cemetery in London) confirmed that the plague was caused by...
 - a) malaria
 - b) the bacterium Yersinia pestis
 - c) influenza
 - d) smallpox
9. ☆ Archaeologists most often link graves aligned east–west in early medieval cemeteries with...
 - a) ship burials
 - b) cremation
 - c) fire worship
 - d) Christian burial rites

PHILOSOPHY

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to halls full of learned monks and masters. Around 524, in prison before his execution, Boethius wrote "The Consolation of Philosophy." In his treatise on music he distinguished the music of the spheres, human music and instrumental music (*instrumentalis*). The cosmos of that time was pictured as concentric heavenly spheres around a motionless Earth. In the eleventh century Anselm of Canterbury formulated a proof of God's existence later called ontological. St. Albert the Great, who wrote much about plants and animals, is today the patron of natural scientists. His student Thomas Aquinas presented the "five ways" – arguments for the existence of God. In the dispute over universals, nominalists claimed that general concepts are just names, and only individual things exist. Ockham's razor advises us not to multiply entities beyond necessity. With his impetus theory, John Buridan explained why a thrown stone keeps flying after it is released. Universities practiced the *disputatio* – a public debate with arguments for and against a thesis. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 In his treatise "De institutione musica," Boethius distinguished the music of the spheres (mundana), the music of the human being (humana) and...
 - a) military music
 - b) folk music
 - c) instrumental music (instrumentalis)
 - d) electronic music
2. 👁 The medieval picture of the cosmos, based on Aristotle and Ptolemy, showed the world as...
 - a) a flat disk on a turtle's back
 - b) an endless void with no center
 - c) a spiral of galaxies
 - d) concentric heavenly spheres around a motionless Earth
3. 📖 In the debate over universals, nominalists argued that general concepts (such as "human") are...
 - a) just names, and only individual things exist
 - b) independent beings in heaven
 - c) sets of numbers
 - d) inventions of poets
4. ✂ The principle known as "Ockham's razor" advises us to...
 - a) trust only authorities
 - b) not multiply entities beyond necessity
 - c) always choose the most complex explanation
 - d) avoid using logic in theology
5. 🚀 Jean Buridan's impetus theory (14th century) tried to explain why...
 - a) bodies always fall straight down
 - b) a thrown stone keeps flying after it is released
 - c) water boils faster in the mountains
 - d) the Moon shines with reflected light
6. 🗣 At medieval universities, a "disputatio" was...
 - a) an exam in calligraphy and handwriting
 - b) a fencing competition between students
 - c) a prayer said before each lecture
 - d) a public debate with arguments for and against a thesis
7. 💖 In prison, before his execution (c. 524), Boethius wrote a work on happiness and the meaning of suffering. What is its title?
 - a) "The Consolation of Philosophy"
 - b) "The City of God"
 - c) "Confessions"
 - d) "Summa Theologiae"
8. 🌿 St. Albert the Great (13th century), teacher of Thomas Aquinas, wrote a great deal about plants, animals and minerals. Whose patron saint is he today?
 - a) Sailors
 - b) Natural scientists
 - c) Miners
 - d) Musicians
9. ☆ The "Five Ways" of St. Thomas Aquinas are...
 - a) pilgrimage routes to Rome
 - b) rules of Latin grammar
 - c) rules of a religious order
 - d) arguments for the existence of God

HISTORY

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, the age of knights and chroniclers. In 966 Mieszko I was baptized. At the beginning of the twelfth century the first chronicle of Poland's history was written in Latin, and it is attributed to Gallus Anonymus. In 1066 the Normans conquered England, and the Norman conquest of England is shown on the Bayeux Tapestry, about 70 meters long. In the twelfth century Peter Abelard described his misfortunes and his love for Heloise. In 1215 Magna Carta limited the power of the King of England over his subjects. The heavy knight's cavalry charge was built around the lance. From 1347 to 1351 the Black Death killed between one third and one half of Europe's people. The Union of Krewo in 1385 joined Poland with the Grand Duchy of Lithuania. In 1410, tradition says, the knights sang "Bogurodzica," the Mother of God hymn, before the Battle of Grunwald. From the baptism to Grunwald, 444 years passed. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🏳️ According to tradition, which song did the Polish knights sing before the Battle of Grunwald (1410)?
 - a) "Dąbrowski's Mazurka"
 - b) "Rota" (The Oath)
 - c) "Bogurodzica" (Mother of God)
 - d) "Gaude Mater Polonia"
2. 👁️ The Bayeux Tapestry, an embroidered strip about 70 m long, depicts...
 - a) the voyage of Columbus
 - b) the siege of Constantinople in 1453
 - c) the coronation of Charlemagne
 - d) the Norman conquest of England in 1066
3. 📖 Who is credited with the first chronicle of Polish history, written in Latin in the early 12th century?
 - a) Jan Długosz
 - b) Mikołaj Rej
 - c) Gallus Anonymus
 - d) Wincenty Kadłubek
4. ➡️ The baptism of Mieszko I took place in 966, and the Battle of Grunwald in 1410. How many years separate these events?

5. ⚔️ Which weapon was the basis of a heavily armored knight's mounted charge?
 - a) The lance
 - b) The sling
 - c) The crossbow
 - d) The boomerang
6. 🤝 With which state was Poland joined by the Union of Krewa (1385)?
 - a) The Grand Duchy of Lithuania
 - b) Bohemia
 - c) Hungary
 - d) The Teutonic Order
7. 💖 Peter Abelard described his misfortunes in the autobiographical letter "Historia calamitatum." What was the name of his beloved?
 - a) Beatrice
 - b) Laura
 - c) Heloise
 - d) Isolde
8. 🍀 What share of Europe's population did the Black Death kill in 1347–1351?
 - a) Between one third and one half
 - b) About 5%
 - c) About 1%
 - d) Almost 95%
9. ⭐ In which year was Mieszko I baptized?
 - a) 1410
 - b) 1000
 - c) 1025
 - d) 966

1 c 2 d 3 c 4 a 5 a 6 a 7 c 8 a 9 d

A N S W E R S

LINGUISTICS

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages and listen to the languages of that time. The scholars and clergy of Western Europe spoke to one another in Latin, and Hebrew, Greek and Latin were held to be the three "sacred" languages of the inscription on Christ's cross. Copyists wrote with goose quills on parchment made of treated animal skin. In the ninth century Constantine, known as Cyril, created a script for the Slavs called Glagolitic. The Vikings used the Younger Futhark, which has 16 characters. Guido of Arezzo named the steps of the scale with the syllables ut, re, mi, taken from the hymn to St. John the Baptist "Ut queant laxis." The Bull of Gniezno of 1136 is the "golden bull of the Polish language" because it contains about 400 Polish place and personal names. The troubadours sang of courtly love in Provençal (Occitan). We took star names such as Aldebaran and Betelgeuse from Arabic. In the fifteenth century "The Imitation of Christ" was written in Latin. Remember these details – I'll ask you about them in a moment.

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1. 🎵 Guido of Arezzo named the steps of the scale with the syllables ut, re, mi, fa, sol, la. Where did he take them from?
 - a) From the names of the nine Muses
 - b) From the opening of the Lord's Prayer
 - c) From the hymn to St. John the Baptist "Ut queant laxis"
 - d) From the letters of the Greek alphabet
2. 👁 What is the name of the script created in the 9th century by Constantine (Cyril) to write the Slavic language?
 - a) Glagolitic
 - b) Cyrillic
 - c) Futhark
 - d) Gothic script
3. 📜 The Bull of Gniezno of 1136 is called the "golden bull of the Polish language" because it...
 - a) is the first book printed in Polish
 - b) contains about 400 Polish place and personal names
 - c) contains the first Polish–Latin dictionary
 - d) was written in gold ink
4. 🏰 How many characters does the Younger Futhark, the runic alphabet of the Viking Age, have?

5. ✂ Medieval scribes wrote with goose quills on parchment, which was made from...
 - a) birch bark
 - b) papyrus
 - c) treated animal skin
 - d) palm leaves
6. 🗣 What language did scholars and clergy of medieval Western Europe use to communicate?
 - a) French
 - b) Greek
 - c) German
 - d) Latin
7. 💖 In what language was the spiritual work "The Imitation of Christ" (15th century), one of the most translated books in the world, written?
 - a) English
 - b) Latin
 - c) Greek
 - d) Polish
8. 🌿 We received the star names Aldebaran, Betelgeuse and Algol through medieval science from...
 - a) Hungarian
 - b) Arabic
 - c) Old Norse
 - d) Chinese
9. ☆ Which languages were regarded in the Middle Ages as the three "sacred" languages of the inscription on Christ's cross?
 - a) Greek, Egyptian and Syriac
 - b) Latin, Gothic and Celtic
 - c) Hebrew, Greek and Latin
 - d) Aramaic, Arabic and Persian

LITERARY STUDIES

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages and open some old manuscripts. They were adorned with illuminations – decorative paintings, initials and borders. Jongleurs recited "The Song of Roland," which belongs to the genre of the *chanson de geste*, the song of deeds. Bestiaries described animals and added a moral or religious lesson to each one. The oldest known sentence in Polish was written down in the Book of Henryków. "The Divine Comedy" was written by Dante Alighieri; the poem has 3 parts, or canticles: Hell, Purgatory and Paradise. In Boccaccio's "Decameron" ten young people flee Florence to escape the plague of 1348. In Chaucer's "Canterbury Tales" the stories are told by pilgrims traveling to Canterbury. The fifteenth century brought the "Holy Cross Lament" – a moving monologue of the Mother of God standing beneath the cross – and "The Conversation of Master Polikarp with Death," a work about death, which comes to everyone. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🎵 To which genre does "The Song of Roland," recited and sung by jongleurs, belong?
 - a) Chanson de geste (song of deeds)
 - b) Epistolary novel
 - c) Liturgical drama
 - d) Sonnet
2. 🖼️ Illuminations in medieval manuscripts are...
 - a) decorative paintings, initials and borders
 - b) wax seals
 - c) watermarks in paper
 - d) candles for reading at night
3. 📖 In which document was the oldest known sentence in Polish written down?
 - a) The Holy Cross Sermons
 - b) The Bull of Gniezno
 - c) The Book of Henryków
 - d) The Chronicle of Gallus Anonymus
4. 📖 How many parts (canticles) make up "The Divine Comedy"?

5. 🚶 In Chaucer's "Canterbury Tales," the stories are told by...
 - a) pilgrims traveling to Canterbury
 - b) monks shut in a monastery
 - c) sailors during a storm
 - d) knights during a siege
6. 🧑 In Boccaccio's "Decameron," ten young people tell each other stories because they have fled Florence to escape...
 - a) a flood
 - b) a war with the Turks
 - c) the plague of 1348
 - d) an earthquake
7. 💖 The "Holy Cross Lament" (15th century), a gem of medieval Polish poetry, is a moving monologue of...
 - a) a dying king on his deathbed
 - b) a penniless wandering student
 - c) a knight on the eve of a great battle
 - d) the Mother of God standing beneath the cross
8. 🌿 Medieval bestiaries described real and fantastic animals, adding to each one...
 - a) a breeding pedigree
 - b) a recipe
 - c) a market price
 - d) a moral or religious lesson
9. ⭐ What is the Polish work "The Conversation of Master Polycarp with Death" (15th century) about?
 - a) A knight's love
 - b) The building of a cathedral
 - c) A crusade
 - d) Death, which comes to everyone

CULTURE AND RELIGION

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, when faith set the rhythm of the day. The Rule of St. Benedict is summed up in the motto "Ora et labora." Churches rang with Gregorian chant – singing that was monophonic, in Latin, with no instrumental accompaniment. In 997 St. Adalbert died a martyr's death on a mission among the Prussians, and in the year 1000 Emperor Otto III met Bolesław the Brave at the Congress of Gniezno. The Great Schism of 1054 split the Church into Western (Latin) and Eastern (Orthodox). In the twelfth century St. Hildegard of Bingen wrote about the healing properties of plants. In his "Canticle of the Sun" St. Francis of Assisi praises God for the sun, water and all creation. In the thirteenth century the hymn "Gaude Mater Polonia" was written in honor of St. Stanislaus. Pilgrims to Santiago de Compostela pinned on a scallop shell, and stained-glass windows filled Gothic cathedrals with colored light. Remember these details – I'll ask you about them in a moment.

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1. 🎵 **Gregorian chant is singing that is...**
 - a) improvised on ritual drums
 - b) monophonic, in Latin, with no instrumental accompaniment
 - c) secular, in vernacular languages, performed by troubadours
 - d) polyphonic, accompanied by the organ
2. 👁 **Which feature most fills the interiors of Gothic cathedrals with colored light?**
 - a) Tapestries
 - b) Ceiling frescoes
 - c) Floor mosaics
 - d) Stained-glass windows
3. 📖 **Which motto sums up the Rule of St. Benedict?**
 - a) "Ora et labora"
 - b) "Memento mori"
 - c) "Carpe diem"
 - d) "Veni, vidi, vici"
4. 🏰 **In which year did Emperor Otto III meet Bolesław the Brave at the Congress of Gniezno?**

5. ✨ **What badge did pilgrims walking to Santiago de Compostela pin to their clothes?**
 - a) A rose
 - b) A key
 - c) A scallop shell
 - d) A horseshoe
6. 🗿 **The Great East–West Schism of 1054 split the Church into...**
 - a) the Roman and Coptic Churches
 - b) Catholics and Protestants
 - c) mendicant and military orders
 - d) Western (Latin) and Eastern (Orthodox)
7. 💖 **In the "Canticle of the Sun," St. Francis of Assisi...**
 - a) sets out the calendar of feasts for the year
 - b) praises God for the sun, water and all creation
 - c) writes down the rule of his order
 - d) describes a crusade to the Holy Land
8. 🌿 **In her works "Physica" and "Causae et curae," St. Hildegard of Bingen (12th century) wrote about, among other things,...**
 - a) the tactics of imperial armies
 - b) the construction of sailing ships
 - c) the healing properties of plants
 - d) the city law of Mainz
9. ⭐ **St. Adalbert was martyred in 997 during...**
 - a) a mission among the Prussians
 - b) the siege of Kraków
 - c) a crusade
 - d) a journey to Rome

1 b 2. d 3. a 4. 1000 5. c 6. d 7. b 8. c 9. a

A N S W E R S

ECONOMICS AND FINANCE

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to the busy market square of a merchant town. A bugle call sounded from the tower, used for marking the time of day and raising alarms. Gdańsk and Toruń belonged to the Hanseatic League – a union of merchant towns on the Baltic and North Seas. The typical freighter of Hanseatic merchants was the cog. Along the Silk Road came silk from Asia, and also spices such as pepper and cinnamon. Coins were struck by hand: the minter hit with a hammer a die placed on a metal blank. Italian money changers sat at benches, and the word "bank" comes from "banco" – the bench where a money changer sat. The Church banned usury, that is lending money at interest, and the Franciscans took a vow of poverty, giving up their property. In 1494 Luca Pacioli described double-entry bookkeeping, in which every transaction is recorded twice: as a debit and a credit. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🏰 In a medieval town, the bugle call played from a church or town-hall tower was used, among other things, for...
 - a) summoning knights to a tournament
 - b) announcing bread and meat prices
 - c) marking the time of day and raising alarms
 - d) advertising goods at the market
2. 🚢 Which ship was the typical cargo vessel of Hanseatic merchants?
 - a) The trireme
 - b) The galleon
 - c) The cog
 - d) The longship
3. 📖 The word "bank" comes from the Italian "banco," meaning...
 - a) a silver coin from Florence
 - b) a merchant's thick account book
 - c) an underground vault in a palace
 - d) the bench where a money changer sat
4. 📊 Double-entry bookkeeping, described in 1494 by Luca Pacioli, means that every transaction is recorded...
 - a) only once, at the end of the year
 - b) twice: as a debit and a credit
 - c) only in gold
 - d) in two different languages
5. 🔨 Medieval coins were struck by hand: the moneyer hit with a hammer...
 - a) clay tablets
 - b) wooden blocks with paint
 - c) hot wax
 - d) a die placed on a metal blank
6. 🏰 What was the name of the league of merchant towns on the Baltic and North Sea that included Gdańsk and Toruń?
 - a) A confederation
 - b) A guild
 - c) The Lombard League
 - d) The Hanseatic League
7. 💜 Besides chastity and obedience, which vow did Franciscans take when giving up their possessions?
 - a) Poverty
 - b) Wandering
 - c) Fasting
 - d) Silence
8. 🌿 Besides silk, the Silk Road carried from Asia to Europe...
 - a) potatoes
 - b) spices such as pepper and cinnamon
 - c) tobacco
 - d) cocoa
9. ⭐ In the Middle Ages, the Church forbade Christians to practice usury, that is,...
 - a) trading on Sunday
 - b) playing dice
 - c) lending money at interest
 - d) wearing gold

PSYCHOLOGY



5. Middle Ages

500–1500 AD

Let's travel back to the Middle Ages and look into the human soul of that time. Medicine relied on humoral theory, which recognized 4 humors, or body fluids. From them came four temperaments: sanguine, choleric, phlegmatic and melancholic. Melancholy was explained by an excess of black bile, and doctors prescribed music for it. The wheel of Fortune motif showed the changeability of fate: rise and fall. The code of chivalry demanded protecting the weak, loyalty and generosity, and among the seven deadly sins pride came first. The mandrake was considered a magical plant because it has a root shaped like a human. In 1374 a dancing mania broke out along the Rhine: crowds danced for hours until exhausted. In the fifteenth century ars moriendi handbooks taught how to prepare well for death. Remember these details – I'll ask you about them in a moment.

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1. 🎵 **Medieval doctors recommended music for melancholy, which humoral theory explained as an excess of...**
 - a) black bile
 - b) blood
 - c) phlegm
 - d) yellow bile
2. 👁 **The "Wheel of Fortune" motif in medieval art depicted...**
 - a) the cycle of the four seasons
 - b) the changeability of fate: rise and fall
 - c) the work of a mill wheel
 - d) the rose window of a Gothic cathedral
3. 🗨 **The four temperaments of humoral theory are sanguine, choleric, phlegmatic and...**
 - a) melancholic
 - b) stoic
 - c) optimistic
 - d) introverted
4. 📖 **How many humors (bodily fluids) did humoral theory, the basis of medieval medicine, recognize?**

5. 🎭 **In 1374, a "dancing mania" broke out along the Rhine. What was it?**
 - a) knights held dance tournaments
 - b) everyone learned to swim
 - c) people stopped walking
 - d) crowds danced for hours until exhausted
6. 🧑 **The medieval code of chivalry required, among other things,...**
 - a) living in solitude as a hermit
 - b) never learning to read or write
 - c) protecting the weak, loyalty and generosity
 - d) avoiding all fighting and conflict
7. 💖 **Which of these is one of the seven deadly sins?**
 - a) Ambition
 - b) Pride
 - c) Curiosity
 - d) Shyness
8. 🌿 **In the Middle Ages, the mandrake was considered a magical plant because it...**
 - a) glows in the dark like a lantern
 - b) blooms only once every hundred years
 - c) grows only on the graves of kings
 - d) has a root shaped like a human
9. ⭐ **The medieval "ars moriendi" handbooks (15th century) taught...**
 - a) how to run a business
 - b) how to build castles
 - c) how to raise falcons
 - d) how to prepare well for death

1. a 2. b 3. a 4. d 5. c 6. d 7. b 8. d 9. d

ANSWERS

EDUCATION



5. Middle Ages

500–1500 AD

Let's travel back to the Middle Ages, to school benches and university halls. In the early Middle Ages cathedral and monastery schools trained mostly future clergy. In the arts faculty students first learned the trivium: grammar, rhetoric and dialectic (logic), and then the quadrivium, which included arithmetic, geometry, astronomy and music. The core of teaching was the lectio – the reading and explaining of a text by a master. The school of Salerno was famous for teaching medicine. On "T and O" maps, Jerusalem was placed at the very center of the world. A boy of knightly birth began as a page and then became a squire. In 1364 King Casimir the Great founded the university in Kraków, and in 1400 the Academy was renewed – 36 years passed between those dates. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🏰 Which subject belonged to the quadrivium, the four mathematical arts taught at medieval universities?
 - a) Dialectic
 - b) Music
 - c) Rhetoric
 - d) Grammar
2. 🗺️ At the very center of medieval "T and O" world maps was...
 - a) Rome
 - b) Jerusalem
 - c) Paris
 - d) Constantinople
3. 📖 The trivium, the three arts of language, comprised grammar, rhetoric and...
 - a) astronomy
 - b) medicine
 - c) dialectic (logic)
 - d) music
4. 🏰 The Kraków Academy was founded in 1364 and restored in 1400. How many years passed between these dates?

5. 🏰 A boy of knightly birth began his training as a page and then became a...
 - a) merchant
 - b) abbot
 - c) village head
 - d) squire
6. 👑 Which king founded the university in Kraków in 1364?
 - a) Louis of Hungary
 - b) Bolesław the Brave
 - c) Władysław Jagiełło
 - d) Casimir the Great
7. ❤️ At a medieval university, the "lectio" was...
 - a) communal prayer in the chapel
 - b) students working in the university garden
 - c) the reading and explaining of a text by a master
 - d) morning fencing practice
8. 🌿 The school of Salerno (from the 10th–11th centuries) was famous throughout Europe for teaching...
 - a) medicine
 - b) navigation
 - c) architecture
 - d) canon law
9. ☆ In the early Middle Ages, cathedral and monastery schools mainly trained...
 - a) merchants
 - b) future clergy
 - c) soldiers
 - d) peasants

1. b 2. b 3. c 4. 365. d 6. d 7. c 8. a 9. b

A N S W E R S

SOCIOLOGY

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages and see how people lived. Society was divided into three estates: the oratores, those who pray, the bellatores, those who fight, and the laboratores – those who work. In the feudal bond a lord granted a vassal a fief, and the vassal was loyal to him and served him in arms. A coat of arms, the family's mark, was placed above all on the shield. A serf had to work a set number of days on the lord's fields. In the thirteenth century founding new villages under German law often went hand in hand with clearing forests for farmland. In towns, guilds brought together craftsmen of a single trade. From the twelfth century in the Low Countries, the Beguines lived devoutly in communities without vows. People were entertained by wandering musicians and jugglers, called *igrcy* in old Poland. In 1300 Pope Boniface VIII proclaimed the first Holy Year. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 What were wandering musicians, jugglers and comedians called in old Poland?
 - a) Minstrels
 - b) Igrcy
 - c) Troubadours
 - d) Goliards
2. 🛡️ A knight's coat of arms, the emblem of his family, was displayed above all on his...
 - a) great helm, on the inside
 - b) horse's horseshoe
 - c) shield
 - d) spur
3. 📖 Medieval society was divided into three estates: oratores (those who pray), bellatores (those who fight) and...
 - a) laboratores (those who work)
 - b) navigatores (sailors)
 - c) doctores (scholars)
 - d) mercatores (merchants)
4. ⚖️ In a feudal relationship, the lord granted the vassal a fief, and in return the vassal...
 - a) paid him a land tax
 - b) gave him his children
 - c) was loyal to him and served him in arms
 - d) became his judge
5. 🧑‍🌾 A peasant bound to serfdom (corvée) had to...
 - a) work a set number of days on the lord's fields
 - b) pay all his dues only in gold
 - c) serve the lord as an armored knight
 - d) move to the town and live there
6. 🧑‍🔧 Guilds in medieval towns brought together...
 - a) overseas merchants
 - b) clergy
 - c) craftsmen of a single trade
 - d) all the town's inhabitants
7. 💜 The Beguines, a movement popular in the Low Countries from the 12th century, were women who...
 - a) were traveling actresses at fairs
 - b) were knights in the Teutonic Order
 - c) lived devoutly in communities without vows
 - d) ran banks in the towns of Flanders
8. 🌿 Founding new villages under German law in the 13th century often involved...
 - a) clearing forests for farmland
 - b) planting vineyards in the mountains
 - c) draining seas
 - d) building gold mines
9. 🌟 In which year did Pope Boniface VIII proclaim the first Holy Year (jubilee)?
 - a) 1492
 - b) 1410
 - c) 1300
 - d) 1054

LAW

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to courtrooms and royal chanceries. In the eleventh century the Russkaya Pravda was written down, the oldest law code of Kievan Rus. Canon law is the law of the Church. A seal hung from a document served to confirm that the document was authentic. Heralds announced the ruler's orders to the sound of trumpets. Trial by combat served to settle a dispute by a fight between the parties, since people believed God would help the one who was right. In Poland the right to mine salt belonged to the king. The Privilege of Koszyce of 1374 freed the nobility from most taxes in exchange for the throne for one of Louis's daughters. The privilege "neminem captivabimus" of 1430/1433 guaranteed personal freedom from arrest without a court verdict. Finally, the Peace of Toruń in 1466, ending the Thirteen Years' War, meant that Poland regained Gdańsk Pomerania. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏰 At tournaments and ceremonies, heralds announced the ruler's decrees to the sound of...
 - a) trumpets
 - b) church bells
 - c) the organ
 - d) war drums
2. 👁 A seal attached to a medieval document served to...
 - a) count its pages
 - b) decorate it
 - c) protect it from moisture
 - d) confirm that the document was authentic
3. 📜 What is the oldest law code of Kievan Rus' (11th century) called?
 - a) Magna Carta
 - b) The Sachsenspiegel
 - c) Russkaya Pravda
 - d) The Code of Justinian
4. 🏰 The Privilege of Koszyce (1374) exempted the nobility from most taxes in exchange for...
 - a) Lithuania accepting baptism
 - b) the nobles joining a crusade
 - c) the throne for one of Louis's daughters
 - d) building new royal castles
5. ⚔ In medieval law, trial by combat was used to...
 - a) mark the boundaries of fields
 - b) settle a dispute by a fight between the parties
 - c) test the writing skills of clerks
 - d) choose a new king for the realm
6. 🏰 Under the Peace of Toruń (1466), which ended the Thirteen Years' War, Poland regained...
 - a) Silesia
 - b) Red Ruthenia
 - c) Masovia
 - d) Gdańsk Pomerania
7. ❤ The nobles' privilege "neminem captivabimus" (1430/1433) guaranteed...
 - a) exemption of noble sons from schooling
 - b) a vote for the nobility in electing the pope
 - c) personal freedom from arrest without a court verdict
 - d) the right of every noble to mint coins
8. 🌿 In medieval Poland, the right to mine salt (the salt regale) belonged to...
 - a) every peasant
 - b) the king
 - c) the butchers' guilds
 - d) the mendicant orders
9. ☆ Canon law is the law of...
 - a) merchants
 - b) the military
 - c) the sea
 - d) the Church

OTHER (HUMANITIES)

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, an era whose name – medium aevum – was given only by Renaissance humanists. Its conventional boundaries are the fall of the Western Roman Empire in 476 and the discovery of America in 1492; 1016 years separate them. In 1154 al-Idrisi drew a world map for Roger II with south at the top. Emperor Frederick II wrote a treatise on hunting with birds of prey, that is on falconry. In the thirteenth century Marco Polo reached the court of Kublai Khan. The Knights of the Round Table searched for the Holy Grail – the cup from the Last Supper. Lay people used books of hours for personal prayer throughout the day. The French played la soule, a game seen as the ancestor of soccer and rugby. In the fifteenth century Nicholas of Radom became one of the first Polish composers known by name – a composer of polyphonic music. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Nicholas of Radom (15th century) is one of the earliest Poles known by name to work as...**
 - a) a royal astronomer
 - b) a builder of cathedrals
 - c) a mapmaker
 - d) a composer of polyphonic music
2. 🗺️ **The world map of al-Idrisi, made in 1154 for King Roger II of Sicily, was oriented with...**
 - a) west at the top
 - b) north at the top
 - c) east at the top
 - d) south at the top
3. 📖 **The name "Middle Ages" (Latin *medium aevum*) was given to this era by...**
 - a) the knights themselves
 - b) ancient historians
 - c) Renaissance humanists
 - d) 6th-century monks
4. 📅 **How many years separate the fall of the Western Roman Empire (476) from Columbus's discovery of America (1492)?**

5. 🏹 **The medieval French team game "la soule" is considered the ancestor of...**
 - a) chess
 - b) fencing
 - c) soccer and rugby
 - d) archery
6. 🧑 **To the court of which ruler did the Venetian merchant Marco Polo travel in the 13th century?**
 - a) Saladin
 - b) Timur
 - c) Charlemagne
 - d) Kublai Khan
7. 💖 **Books of hours (such as the "Très Riches Heures du Duc de Berry") were used by lay people for...**
 - a) keeping accounts
 - b) recording court verdicts
 - c) learning fencing
 - d) personal prayer throughout the day
8. 🍃 **Emperor Frederick II (13th century) wrote a famous treatise on hunting with...**
 - a) bear traps
 - b) fishing nets
 - c) birds of prey (falconry)
 - d) hunting dogs
9. ⭐ **In Arthurian legend, the Knights of the Round Table sought the Holy Grail, which was...**
 - a) Arthur's crown
 - b) the cup from the Last Supper
 - c) the sword Excalibur
 - d) the key to paradise

ARCHITECTURE AND URBAN PLANNING

**5. Middle Ages**

500–1500 AD

Let's travel back to the Middle Ages, to the building site of a Gothic cathedral. Such a church, full of light and reaching upward, was meant to symbolize the Heavenly Jerusalem. Its hallmark is the pointed arch. The flying buttresses outside transfer the vault's thrust to outer piers, so the walls could hold huge windows. Above the entrance glows the rose window – a large round window with stone tracery and stained glass. Stones were lifted by treadwheel cranes powered by people walking inside a large wheel. The largest instrument in the church was the pipe organ. Cologne Cathedral was begun in 1248 and finished in 1880 – after 632 years. In a monastery, the cloister served for quiet prayer, reading and meditation. The Teutonic Knights' castles, including Malbork, were built of brick. And founding a town under Magdeburg law meant self-government with a council and a regular street plan around a market square. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 What was the largest instrument in Gothic churches?
 - a) The kettledrum
 - b) The lute
 - c) The pipe organ
 - d) The harp
2. 🏰 Flying buttresses on the outside of Gothic cathedrals serve to...
 - a) vent smoke from fireplaces
 - b) transfer the vault's thrust to outer piers
 - c) support the bells
 - d) decorate with no function
3. 📖 In Gothic architecture, a rose window is...
 - a) a churchyard garden
 - b) a stone floor shaped like a rose
 - c) a large round window with stone tracery and stained glass
 - d) a small tower with a bell
4. 🏰 Construction of Cologne Cathedral began in 1248 and was completed only in 1880. How many years passed?

5. ⚙️ Medieval treadwheel cranes were powered by...
 - a) the wind
 - b) people walking inside a large wheel
 - c) water wheels
 - d) oxen pulling a plow
6. 🏰 Founding a town under Magdeburg law meant, among other things,...
 - a) placing the whole town under a monastery
 - b) a total ban on trade within the walls
 - c) self-government with a council and a regular street plan
 - d) an obligation to build a sea port
7. 💖 The arcaded cloister (garth) in a monastery served mainly for...
 - a) storing grain
 - b) knightly training
 - c) quiet prayer, reading and meditation
 - d) trade
8. 🍃 What material was used to build most Teutonic castles, including Malbork?
 - a) Wood
 - b) Marble
 - c) Brick
 - d) Granite
9. ⭐ A Gothic cathedral, full of light and soaring upward, was meant to symbolize...
 - a) the Heavenly Jerusalem
 - b) the power of rich merchants
 - c) an impregnable fortress
 - d) a victorious emperor's palace

1 c 2. b 3. c 4. 6325. b 6. c 7. c 8. c 9. a

A N S W E R S



5. Middle Ages

500–1500 AD

Let's travel back to the Middle Ages, to a world of gears and clever machines. The word "automaton" comes from the Greek "automatos," acting of itself. Around 1200 al-Jazari described a boat-shaped automaton with musicians that played by itself, powered by water. Mills were driven by a water wheel. In monasteries the hours of prayer were measured with water clocks, hourglasses and candles marked with a scale. At the end of the thirteenth century the first mechanical clocks kept time with a falling weight and an escapement. Tower clocks meant townspeople could agree on times for work, markets and meetings. The dial of the Prague Orloj of 1410 shows the positions of the Sun and Moon in the sky. In 1269 Peter Peregrinus described how a magnet has two poles, north and south. And Nicole Oresme compared the universe to a great clock wound up by God. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Al-Jazari (12th/13th century) described a boat-shaped automaton with musicians that...**
 - a) needed four people to crank it
 - b) played by itself, powered by water
 - c) was merely decorative
 - d) was controlled by electricity
2. 👁 **The dial of Prague's astronomical clock (Orloj, 1410) shows, among other things,...**
 - a) a list of Bohemian kings
 - b) market prices
 - c) the positions of the Sun and Moon in the sky
 - d) the weather forecast
3. 🗨 **The word "automaton" comes from the Greek "automatos," meaning...**
 - a) hand-cranked
 - b) fast as the wind
 - c) made of copper
 - d) acting of itself
4. ⚙ **The first mechanical clocks (late 13th century) measured time thanks to...**
 - a) electric current in a coil
 - b) sunlight reflected from a mirror
 - c) a falling weight and an escapement
 - d) steam pressure in a boiler
5. ⚡ **What is the device called in which falling or flowing water turns a wheel that drives a mill?**
 - a) A spinning wheel
 - b) A water wheel
 - c) A treadmill
 - d) A windmill
6. 🏰 **Clocks on town-hall and church towers in the 14th–15th centuries meant that townspeople...**
 - a) could agree on times for work, markets and meetings
 - b) had to pay a tax for every hour
 - c) never went out after dark again
 - d) stopped attending prayers in church
7. 💖 **In monasteries, the times of prayer (canonical hours) were measured with water clocks, hourglasses and...**
 - a) candles marked with a scale
 - b) a compass
 - c) the telephone
 - d) a balance scale
8. 🌿 **In his "Letter on the Magnet" (1269), Peter Peregrinus of Maricourt described how a magnet has...**
 - a) a pole only at night
 - b) three poles
 - c) only one pole
 - d) two poles, north and south
9. ☆ **Nicole Oresme (14th century) compared the universe to...**
 - a) the tree of life
 - b) an endless river
 - c) a ball of fire
 - d) a great clock wound up by God

COMPUTER SCIENCE AND TELECOMMUNICATIONS

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages and see how news was passed on. Ringing the bells in alarm announced danger, such as fire or invasion. On the towers of the Great Wall of China, signals were sent with smoke by day and fire by night. In the thirteenth century the Mongols carried orders through the "yam" system – a network of stations with fresh horses. Homing pigeons can carry messages because they return to their loft even from far away. Preachers memorized texts with the memory palace method, imagining places for successive thoughts. In scriptoria, books were copied by hand, page by page. The word "algorithm" comes from the name of the Persian mathematician al-Khwarizmi. The Roman date MCDX is simply 1410 – the year of the Battle of Grunwald. And around 1455 Johannes Gutenberg printed the first great book with movable type: the Bible. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Ringling the bells as an alarm told townspeople about...**
 - a) a royal wedding
 - b) danger, such as fire or invasion
 - c) the end of harvest
 - d) the start of a fair
2. 👁 **On the signal towers of the Great Wall of China, messages were sent by...**
 - a) letters on arrows
 - b) smoke by day and fire by night
 - c) sun mirrors
 - d) pigeons
3. 📖 **The word "algorithm" comes from the name of...**
 - a) the Greek philosopher Aristotle
 - b) the merchant Algernon
 - c) the monk Alcuin
 - d) the Persian mathematician al-Khwarizmi
4. 🗓 **Write the Roman date of the Battle of Grunwald, MCDX, in Arabic numerals.**

5. ✂ **In monastery scriptoria, books were reproduced by copying them...**
 - a) with a printing press from the 6th century
 - b) on a typewriter
 - c) using carbon paper
 - d) by hand, page by page
6. 🐾 **In the 13th century, the Mongols relayed orders through the "yam" system, which was...**
 - a) boats on great lakes
 - b) sled dogs on the steppe
 - c) towers with signal mirrors
 - d) a network of stations with fresh horses
7. 💖 **Medieval preachers memorized long texts using the "memory palace" method, which involves...**
 - a) imagining places for successive thoughts
 - b) writing everything on your palm
 - c) singing the text on one note
 - d) repeating the text backward
8. 🕊 **Homing pigeons can carry messages because they...**
 - a) fly only at night
 - b) know the way to every city
 - c) return to their loft even from far away
 - d) understand human speech
9. ☆ **What was the first major book Johannes Gutenberg printed with movable type (c. 1455)?**
 - a) The Quran
 - b) "The Divine Comedy"
 - c) "Pan Tadeusz"
 - d) The Bible

1. b 2. b 3. d 4. 1410 5. d 6. d 7. a 8. c 9. d

A N S W E R S

BIOMEDICAL ENGINEERING

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, when the first aids for the body and eyes appeared. Doctors placed their fingers on the wrist to check the pulse. Around the year 1000 the surgeon al-Zahrawi, known as Albucasis, described and drew about 200 surgical instruments. Hospitals were usually run by religious orders, such as the Hospitallers. The Salerno rules of health advised moderate eating, sleep and good spirits. The patron of doctors is St. Luke, evangelist and physician. Readers used a "reading stone" – a glass or crystal lens that magnified text. At the end of the thirteenth century the first eyeglasses appeared in Italy. Their convex lenses helped people who could not see well up close. Glass for spectacles and stained glass was melted mainly from sand with added ash. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 Which bodily rhythm did medieval doctors examine by placing their fingers on the wrist?
 - a) breathing
 - b) the pulse
 - c) blinking
 - d) walking pace
2. 👁 The first reading glasses appeared at the end of the 13th century in...
 - a) China
 - b) Italy
 - c) Egypt
 - d) Poland
3. 📖 A "reading stone" (lapis lectionis) was...
 - a) a gemstone on a book cover
 - b) a stone for sharpening quills
 - c) a stone tablet with the alphabet
 - d) a glass or crystal lens that magnified text
4. 👓 The convex lenses in the first eyeglasses helped people who...
 - a) had ear and throat trouble
 - b) could not see well up close
 - c) could not tell colors apart
 - d) saw poorly at night
5. 🗺 In his work "Al-Tasrif," the surgeon al-Zahrawi (Abulcasis, c. 1000) described and drew...
 - a) a plan of Córdoba
 - b) siege engines
 - c) about 200 surgical instruments
 - d) a star map
6. 🏰 Medieval hospitals were most often run by...
 - a) blacksmiths' and locksmiths' guilds
 - b) law schools
 - c) royal armies in peacetime
 - d) religious orders, such as the Hospitallers
7. ❤ The Salernitan rules of healthy living ("Regimen sanitatis Salernitanum") recommended, among other things,...
 - a) eating as much meat as possible
 - b) moderate eating, sleep and good spirits
 - c) avoiding water for life
 - d) sleeping only during the day
8. 🍷 Glass for eyeglasses and stained-glass windows was mainly melted from...
 - a) rock salt from Wieliczka
 - b) sand with added ash
 - c) beeswax and tar
 - d) white Italian marble
9. ⭐ Which evangelist, a physician by profession, is the patron saint of doctors?
 - a) St. Mark
 - b) St. John
 - c) St. Matthew
 - d) St. Luke

CHEMICAL ENGINEERING

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to the workshops of alchemists, bell founders and dyers. Alchemy combined experiments with substances with philosophy and religious symbolism. Alchemists sought the philosopher's stone, which was meant to turn metals into gold and prolong life. They used the alembic for distillation. The word "alcohol" comes from the Arabic "al-kuhl," a fine powder for painting eyelids. Around the ninth century gunpowder was invented by the Chinese; it is a mixture of saltpeter, charcoal and sulfur. Bells were cast from an alloy of copper and tin (bell bronze). Soap was boiled from fat and lye from wood ash. And dyers got the color blue from the woad plant. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏰 Church bells were cast from an alloy of...
 - a) iron and carbon
 - b) lead and mercury
 - c) copper and tin (bell bronze)
 - d) gold and silver
2. 👁 What was an alembic, the alchemists' vessel, used for?
 - a) Measuring time
 - b) Weighing gold
 - c) Distillation
 - d) Storing grain
3. 🗨 The word "alcohol" comes from the Arabic "al-kuhl," which originally meant...
 - a) a fine powder for painting eyelids
 - b) a strong red wine
 - c) a sacred fire
 - d) a remedy for coughs
4. ⚗ Gunpowder is a mixture of saltpeter, charcoal and...
 - a) table salt
 - b) sulfur
 - c) sand
 - d) flour
5. ✨ Medieval soap was boiled from fat (tallow or olive oil) and...
 - a) lye from wood ash
 - b) vinegar
 - c) flour
 - d) honey
6. 🧑 Who is credited with inventing gunpowder (c. 9th century)?
 - a) The Vikings
 - b) The Aztecs
 - c) The Chinese
 - d) The Romans
7. ❤ Alchemists sought the "philosopher's stone," which was supposed to...
 - a) turn metals into gold and prolong life
 - b) point north during voyages
 - c) glow at night without fire
 - d) forecast the weather for harvest
8. 🌿 What color did the woad plant give to dyers?
 - a) Black
 - b) Yellow
 - c) Red
 - d) Blue
9. ⭐ Medieval alchemy combined experiments with substances with...
 - a) accounting
 - b) the art of war
 - c) maritime law
 - d) philosophy and religious symbolism

CIVIL ENGINEERING AND TRANSPORT



5. Middle Ages

500–1500 AD

Let's travel back to the Middle Ages and set out on the road. Distances were measured in Roman miles, 1000 double paces. The Silk Road linked China with the Mediterranean, and the word "caravan" comes from Persian. The route "from the Varangians to the Greeks" linked Scandinavia with Byzantium (Constantinople). The Vikings reached North America around the year 1000, and Columbus arrived 492 years later, in 1492. At the end of the 12th century the magnetic compass appeared in Europe. On land, the horse collar allowed a horse to pull a heavy plow without choking. Bridge building was supported with alms because it was seen as a pious work of mercy. And a pilgrimage to Rome, Jerusalem or Santiago was above all penance and a path of spiritual transformation. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏰 The Roman mile, also used in the Middle Ages, was 1,000...
 - a) fathoms
 - b) double paces
 - c) cubits
 - d) feet
2. 🗺️ The Silk Road connected...
 - a) China with the Mediterranean
 - b) Poland with Spain
 - c) Egypt with America
 - d) Scandinavia with Greenland
3. 🗨️ From which language does the word "caravan" come?
 - a) Hungarian
 - b) Persian
 - c) German
 - d) Latin
4. 🚢 The Vikings reached North America around 1000, and Columbus in 1492. How many years later did Columbus arrive?

5. 🐎 The horse collar (a rigid collar), which spread in medieval Europe, allowed...
 - a) a rider to gallop much faster
 - b) oxen to swim across rivers
 - c) donkeys to carry heavy armor
 - d) a horse to pull a heavy plow without choking
6. 🌐 The trade route "from the Varangians to the Greeks" linked Scandinavia with...
 - a) Byzantium (Constantinople)
 - b) Ireland
 - c) Rome
 - d) Córdoba
7. ❤️ For medieval people, a pilgrimage to Rome, Jerusalem or Santiago was above all...
 - a) a profitable trading expedition
 - b) penance and a path of spiritual transformation
 - c) a paid profession for the poor
 - d) a way to escape paying taxes
8. 🌿 When did the magnetic compass, used in navigation, appear in Europe?
 - a) At the end of the 12th century
 - b) In the 17th century
 - c) In the 19th century
 - d) In the 5th century
9. ⭐ In the Middle Ages, bridge building was supported with alms and indulgences because it was considered...
 - a) a pious work of mercy
 - b) a punishment for criminals
 - c) a game for children
 - d) a duty of knights

MATERIALS ENGINEERING

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to forges, glassworks and workshops. Bloomery furnaces could not reach the melting point of iron, so they produced a spongy bloom of iron that had to be forged. A smith hardened a blade by heating it and quickly cooling it in water. Damascus steel takes its name from Damascus. In 1291 the Venetian glassworks were moved to the island of Murano. Colored glass for stained-glass windows was made by adding metal oxides, such as cobalt. Organ pipes were made of an alloy of tin and lead. Paper was made mainly from linen rags, and ink from oak galls and iron sulfate. Gold in the backgrounds of icons and illuminations symbolized God's light and eternity. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎺 **Organ pipes were most often made from an alloy of...**
 - a) silver and gold
 - b) tin and lead
 - c) copper and zinc
 - d) iron and carbon
2. 🌈 **Colored glass for stained-glass windows was made by adding to the molten glass...**
 - a) colored river sand
 - b) fruit and vegetable juices
 - c) metal oxides, such as cobalt
 - d) oak-gall ink
3. 🏰 **Which city gave its name to the famous patterned steel of medieval blades?**
 - a) Damietta
 - b) Toledo
 - c) Damascus
 - d) Aleppo
4. 🔥 **Medieval bloomeries and furnaces did not reach the melting point of iron, so they produced...**
 - a) glass
 - b) a spongy bloom of iron that had to be forged
 - c) liquid iron ready for casting
 - d) pure gold
5. ⚔️ **A blacksmith hardened a steel blade by...**
 - a) burying the blade in snow for the winter
 - b) polishing the blade with wet sand
 - c) heating it and quickly cooling it in water
 - d) soaking the blade in wine for a long time
6. 🏝️ **To which island were the Venetian glassworks moved in 1291?**
 - a) Sicily
 - b) Malta
 - c) Murano
 - d) Crete
7. 💜 **The ink that scribes used to carefully copy books was made, among other things, from...**
 - a) sheep's milk and flour
 - b) tallow soot and linden honey
 - c) wild raspberry juice
 - d) oak galls and iron sulfate
8. 🌿 **In medieval Europe, paper was made mainly from...**
 - a) sugar cane
 - b) bamboo
 - c) palm leaves
 - d) linen rags
9. ⭐ **Gold backgrounds in medieval icons and illuminations symbolized...**
 - a) the summer sun
 - b) merchants' wealth
 - c) God's light and eternity
 - d) the color of the earth

MECHANICAL ENGINEERING



5. Middle Ages

500–1500 AD

Let's travel back to the Middle Ages, to a world of levers, wheels and engines. The word "mechanics" comes from the Greek "mechane," a device or machine. A knight wore mail made of interlinked metal rings. The trebuchet hurled missiles thanks to a heavy counterweight, and its long arm moved very fast because the far end of the lever sweeps a larger arc. A post mill was turned on its post to face its sails into the wind. For using the lord's mill, peasants paid a share of the ground grain (a measure). The spinning wheel, which appeared in the thirteenth century, sped up spinning thread from wool and flax. In organs, air was pumped by bellows worked by people (calcants). Villard de Honnecourt sketched a wheel meant to turn forever without any power. And at the end of the era Leonardo da Vinci kept his notes in mirror writing, from right to left. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎺 In medieval organs, air was pumped into the pipes by...
 - a) electric fans
 - b) waterfalls
 - c) windmills on the roof
 - d) bellows worked by people (calcants)
2. 🏹 The trebuchet, a medieval siege engine, hurled projectiles thanks to...
 - a) twisted ropes of hair
 - b) compressed air
 - c) exploding gunpowder
 - d) a heavy counterweight
3. 🗨 The word "mechanics" comes from the Greek "mechane," meaning...
 - a) wind
 - b) weight
 - c) motion of the planets
 - d) device, machine
4. ⚖ In a trebuchet, the long arm moves very fast because...
 - a) a magnet in the tower pulls the projectile
 - b) the arm is hollow inside
 - c) the far end of the lever sweeps a larger arc
 - d) the counterweight is lighter than the projectile
5. 🗡 What was chain mail, the medieval armor, made of?
 - a) Wooden slats
 - b) A single forged steel plate
 - c) Interlinked metal rings
 - d) Hardened leather
6. 🧑 Peasants had to use mills belonging to the lord or monastery, paying with...
 - a) a share of the ground grain (a measure)
 - b) prayer
 - c) silver only
 - d) tax relief
7. 💜 How did Leonardo da Vinci write his notes full of machine sketches?
 - a) Only in drawings, without words
 - b) In a numerical cipher
 - c) In Greek
 - d) In mirror writing, from right to left
8. 🍃 A post mill was turned as a whole on its post in order to...
 - a) face its sails into the wind
 - b) change its color
 - c) move it to another field
 - d) shelter it from the rain
9. ☆ Villard de Honnecourt (13th century) sketched in his notebook a wheel that was supposed to...
 - a) pump seawater into the city
 - b) turn forever without any power
 - c) measure prayer times in a monastery
 - d) fly over the battlefield



5. Middle Ages

500–1500 AD

Let's travel back to the Middle Ages, to mines, forges and mills. The rhythmic strokes of hammers in forges and fulling mills were driven by a water wheel. If a mill wheel makes 10 turns a minute, in one hour it makes 600. The Cistercian rule stressed that monks should work with their own hands, e.g., in the fields. Miners broke hard rock by fire-setting: they heated the rock and doused it with water. A "župa" was a salt mine and its administration – hence the Kraków salt works. Around 1300 King Wenceslaus II issued the mining law for the silver mines of Kutná Hora. The patron saint of miners is St. Barbara. From around 1400 Kraków drew water from wells and wooden water pipes. And in 1306 Edward I banned burning coal in London because the smoke polluted the air. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The rhythmic blows of hammers in medieval forges and fulling mills were powered by...
 - a) a water wheel
 - b) steam
 - c) windmills on the roof
 - d) animals on a treadmill
 2. 👁 Medieval towns, such as Kraków from around 1400, drew water from wells and...
 - a) channels from glaciers
 - b) rain barrels on roofs only
 - c) marble aqueducts
 - d) wooden water pipes
 3. 🗨 The old Polish word "żupa," as in the royal "żupy krakowskie," meant...
 - a) a miners' guild
 - b) a royal forest
 - c) a river port
 - d) a salt mine (and its administration)
 4. ⚙ A mill's water wheel makes 10 turns per minute. How many turns will it make in an hour?

 5. 🔥 Medieval miners broke hard rock using fire-setting, which means they...
 - a) heated the rock and doused it with water
 - b) melted it in a large furnace
 - c) blasted it with dynamite charges
 - d) dissolved it with acid from barrels
 6. 👤 Which king issued, around 1300, the famous mining law for the silver mines of Kutná Hora (*Ius regale montanorum*)?
 - a) Charles IV
 - b) Wenceslaus II
 - c) Bolesław the Brave
 - d) Casimir the Great
 7. 💖 The Cistercian rule stressed that monks should...
 - a) never leave the church building
 - b) live on alms from the faithful alone
 - c) work with their own hands, e.g., in the fields
 - d) engage mainly in long-distance trade
 8. 🍃 In 1306, King Edward I of England banned burning coal in London because...
 - a) the army needed it
 - b) coal was too expensive
 - c) the smoke polluted the air
 - d) coal exploded
 9. ☆ Which saint is the patron of miners?
 - a) St. Hedwig
 - b) St. Agnes
 - c) St. Barbara
 - d) St. Catherine
-

ASTRONOMY

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages and look up at the night sky. In Ptolemy's system the Earth stood at the center of the world. The word "zenith" comes from Arabic. The astrolabe was used for finding the positions of stars and the time, and the quadrant measured a star's altitude above the horizon. The hymn "Ave maris stella" calls Mary the Star of the Sea. The Julian calendar was slightly too long and drifted by 1 day about every 128 years. In the thirteenth century the Alfonsine Tables were made at the request of Alfonso X the Wise, King of Castile. Halley's Comet of 1066 appears on the Bayeux Tapestry as a bad omen for King Harold. And in 1054 Chinese astronomers recorded a "guest star" – a supernova that left behind the Crab Nebula. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The medieval hymn "Ave maris stella" calls Mary the...
 - a) Moon of the East
 - b) Star of the Sea
 - c) Sun of the West
 - d) Queen of the Night
2. 👁 The astrolabe was used for...
 - a) finding the positions of stars and the time
 - b) measuring water depth in a harbor
 - c) weighing coins at the mint
 - d) counting soldiers in a camp
3. 🗨 From which language does the word "zenith" come?
 - a) Greek
 - b) Latin
 - c) Hebrew
 - d) Arabic
4. 📅 The Julian calendar was slightly too long, so it drifted against the seasons by 1 day about every...
 - a) 1,000 years
 - b) 4 years
 - c) 128 years
 - d) 12 years
5. 🧭 The quadrant, an instrument of astronomers and sailors, was used to measure...
 - a) the air temperature at night
 - b) the sea depth beneath a ship
 - c) a star's altitude above the horizon
 - d) a ship's speed at sea
6. 👤 On whose orders were the "Alfonsine Tables" of planetary motion compiled in the 13th century?
 - a) Alfonso X the Wise, King of Castile
 - b) Richard the Lionheart
 - c) Charlemagne
 - d) Casimir the Great
7. 💜 Halley's Comet of 1066 is shown on the Bayeux Tapestry as...
 - a) a fallen moon
 - b) a good sign for sailors
 - c) a bad omen for King Harold
 - d) the Star of Bethlehem
8. 🌿 In 1054, Chinese astronomers recorded a "guest star." It was a supernova that left behind...
 - a) the Big Dipper
 - b) Halley's Comet
 - c) the Crab Nebula
 - d) the Milky Way
9. 🌟 In the Ptolemaic system, accepted in the Middle Ages, the center of the world was...
 - a) the North Star
 - b) the Earth
 - c) Jerusalem on the Moon
 - d) the Sun

1 b 2 a 3 d 4 c 5 c 6 a 7 c 8 c 9 b

ANSWERS

MATHEMATICS



5. Middle Ages

500–1500 AD

Let's travel back to the Middle Ages and count along with its scholars. In the sixth century the monk Dionysius Exiguus proposed counting years from the birth of Christ. On an abacus people calculated by moving pebbles or counters along lines on a board. The digits 0–9 reached Europe through the Arabs, who adopted them from India. The word "algebra" comes from al-Khwarizmi's work, in which "al-jabr" means restoring, that is moving terms across. In 1202 Leonardo of Pisa, called Fibonacci, popularized Arabic numerals and the decimal system in his "Liber Abaci." He illustrated his sequence with a puzzle about breeding rabbits: 1, 1, 2, 3, 5, 8, 13, and then 21. On a monochord, halving the string gives the interval of an octave. The decorations of the Alhambra are famous for repeating geometric patterns. And the number 7 meant fullness and perfection. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 On a monochord, shortening a string by half (a 2:1 ratio) produces the interval of...
 - a) a second
 - b) a fifth
 - c) a third
 - d) an octave
2. 👁 The decorations of the Alhambra palace in Granada (13th–14th centuries) are famous for...
 - a) portraits of kings
 - b) maps of the world
 - c) repeating geometric patterns
 - d) hunting scenes
3. 🗨 The word "algebra" comes from the title of al-Khwarizmi's work, in which "al-jabr" means...
 - a) restoring (moving terms across)
 - b) a lunar calendar
 - c) a right triangle
 - d) the number zero in writing
4. 📖 In his "Liber Abaci" (1202), Leonardo of Pisa (Fibonacci) popularized in Europe...
 - a) differential calculus
 - b) logarithms
 - c) Roman numerals
 - d) Arabic numerals and the decimal system
5. 🧮 On an abacus (counting board), people calculated by moving...
 - a) water between vessels
 - b) rings on their fingers
 - c) a quill across parchment
 - d) pebbles or counters along lines on a board
6. 🌐 The digits we use today (0–9) reached Europe through...
 - a) the Arabs, who adopted them from India
 - b) the Romans
 - c) the Chinese via Russia
 - d) the Vikings
7. ❤ The monk Dionysius Exiguus (6th century) proposed counting years from...
 - a) the founding of Rome
 - b) the Flood
 - c) the birth of Christ
 - d) the reign of Diocletian
8. 🐞 In "Liber Abaci" (1202), Fibonacci illustrated his famous sequence with a puzzle about the breeding of...
 - a) swallows
 - b) bees
 - c) rabbits
 - d) fish
9. ⭐ In medieval symbolism, the number 7 (e.g., seven sacraments, seven gifts of the Holy Spirit) meant...
 - a) poverty
 - b) bad luck
 - c) fullness and perfection
 - d) chaos

OTHER (TECHNOLOGY)

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, an age of small but groundbreaking inventions. In the sixth century silkworm eggs were smuggled from Asia to Byzantium, and silk production began there. Around 1040 in China, Bi Sheng invented movable type of fired clay. In the twelfth century the first paper mill in Europe, learned from the Arabs, appeared in Spain (Xàtiva). Johannes Gutenberg came from Mainz. An hourglass measures time because sand flows through at a steady rate, and sundials on church walls showed the times of services and prayers. A crossbow differs from a bow in that its string is drawn by a mechanism. In processions people played the portative – a small organ carried on a strap. Bells were solemnly blessed and often given names. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎺 A small organ carried on a strap and played in processions is called a...
 - a) barrel organ
 - b) harpsichord
 - c) church positive
 - d) portative
2. 👁 Who invented movable type made of fired clay in China around 1040?
 - a) Genghis Khan
 - b) Marco Polo
 - c) Confucius
 - d) Bi Sheng
3. 🗨 Which city was Johannes Gutenberg from?
 - a) Venice
 - b) Kraków
 - c) Paris
 - d) Mainz
4. ⚖ A sand hourglass measures time because...
 - a) sand slowly evaporates from the glass
 - b) sand flows through at a steady rate
 - c) sand changes color over time
 - d) the glass heats up every hour
5. 🏹 A crossbow differs from a bow in that...
 - a) it shoots stones
 - b) its string is drawn by a mechanism
 - c) it has no string at all
 - d) it is meant only for children
6. 🧑 Where was the first paper mill in Europe (12th century), adopted from the Arabs?
 - a) In Sweden (Uppsala)
 - b) In Spain (Xàtiva)
 - c) In Poland (Kraków)
 - d) In England (Oxford)
7. ❤ Sundials carved on the walls of medieval churches showed...
 - a) the age of the building
 - b) the wind direction
 - c) the dates of movable feasts
 - d) the times of services and prayers
8. 🍃 Silk production began in Byzantium in the 6th century, when someone smuggled in from Asia...
 - a) cats
 - b) rice harvests
 - c) silkworm eggs
 - d) cotton seeds
9. ☆ In the Middle Ages, church bells were solemnly blessed and often given...
 - a) guild coats of arms
 - b) serial numbers
 - c) names
 - d) names of cities

BIOLOGY

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages and look at the animals of that time. The word "bestiary" comes from the Latin "bestia," meaning animal. In bestiaries the pelican feeding its chicks with its own blood was a symbol of Christ giving his life for people. According to the theory of spontaneous generation, flies came from rotting meat. The strings of lutes, fiddles and harps were usually made of animal gut. Emperor Frederick II, in his treatise "De arte venandi cum avibus," described and had illustrated the anatomy and behavior of birds. Legend says St. Francis preached a sermon to the birds. A knight's warhorse, the destrier, had to be strong and sturdy to carry an armored rider. In Poland beekeeping in the forest meant collecting honey from hives in hollow trees. And the aurochs, the wild ancestor of cattle, still lived in Poland, in Masovia. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎻 **Strings for medieval lutes, fiddles and harps were most often made from...**
 - a) nylon
 - b) animal gut
 - c) horsehair
 - d) steel wire
2. 👁 **In his treatise "De arte venandi cum avibus," Emperor Frederick II described and had illustrated...**
 - a) the anatomy and behavior of birds
 - b) medicinal plants
 - c) the constellations
 - d) the fish of the Mediterranean
3. 📖 **The word "bestiary" comes from the Latin "bestia," meaning...**
 - a) cage
 - b) hunter
 - c) animal
 - d) sea monster
4. 🦟 **According to the theory of spontaneous generation, widespread in the Middle Ages, flies arose...**
 - a) from rotting meat
 - b) from eggs laid by other flies
 - c) from stardust
 - d) from rain
5. 🐎 **A knight's warhorse (destrier) had to be above all...**
 - a) used only for pulling a plow
 - b) wild and unbroken
 - c) strong and sturdy to carry an armored rider
 - d) as small as possible
6. 🐝 **Forest beekeeping, an important occupation in medieval Poland, involved...**
 - a) burning charcoal in the forest
 - b) catching fish in large nets
 - c) collecting honey from hives in hollow trees
 - d) making wooden beer barrels
7. 💖 **According to a well-known legend, St. Francis of Assisi preached a sermon...**
 - a) to knights' horses
 - b) to the fish in the Tiber
 - c) to the birds
 - d) to lions in Africa
8. 🌿 **The aurochs, the wild ancestor of cattle, still lived in the Middle Ages in places such as...**
 - a) Egypt
 - b) Poland (Masovia)
 - c) England
 - d) Iceland
9. ⭐ **In medieval art, a pelican feeding its chicks with its own blood was a symbol of...**
 - a) war
 - b) greed
 - c) spring
 - d) Christ giving his life for people

CHEMISTRY

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to an alchemist's smoky workshop. Following Aristotle, scholars held that the world is made of four elements: earth, water, air and fire. Alchemists linked metals to heavenly bodies: gold to the Sun, silver to the Moon. The symbol of gold was a circle with a dot in the center, the Sun sign. They wrote their knowledge in symbols and allegories to hide knowledge from the uninitiated. Europe read Jabir ibn Hayyan, called Geber – an Arab alchemist translated into Latin. The word "elixir" comes from Arabic. In an alembic, during distillation, a liquid first evaporates and then condenses. From the thirteenth and fourteenth centuries people knew aqua regia, which can dissolve even gold. Meanwhile in China firecrackers and fireworks went bang thanks to the rapid burning of gunpowder. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🧨 **Chinese firecrackers and fireworks went bang thanks to...**
 - a) water mixed with lime
 - b) magnets hidden inside
 - c) compressed steam in tubes
 - d) the rapid burning of gunpowder
2. 🌀 **What did the alchemical symbol for gold look like?**
 - a) A circle with a dot in the center (the Sun sign)
 - b) A cross inside a circle
 - c) A downward-pointing triangle
 - d) A crescent moon
3. 📖 **From which language does the word "elixir" come?**
 - a) Latin
 - b) Arabic
 - c) Hungarian
 - d) German
4. ⚗️ **Alchemists matched metals with heavenly bodies: gold with the Sun, silver with...**
 - a) Saturn
 - b) Mars
 - c) Venus
 - d) the Moon
5. 🔥 **During distillation in an alembic, the liquid first...**
 - a) instantly bursts into flame
 - b) freezes and cracks into pieces
 - c) evaporates and then condenses
 - d) turns into shiny metal
6. 🧑 **Jabir ibn Hayyan (known in Europe as Geber, 8th century) was...**
 - a) an Arab alchemist translated into Latin
 - b) an 8th-century Byzantine emperor
 - c) a Viking chieftain from Ireland
 - d) a Persian poet from Shiraz
7. 💖 **Alchemists often wrote their recipes in symbols and allegories in order to...**
 - a) hide knowledge from the uninitiated
 - b) decorate their books
 - c) let illiterate people read them
 - d) save ink
8. 🌿 **Aqua regia (royal water), a mixture of acids known to alchemists from the 13th–14th centuries, can dissolve...**
 - a) wood
 - b) gold
 - c) glass
 - d) clay
9. ⭐ **According to medieval science (following Aristotle), the world was made of four elements: earth, water, air and...**
 - a) gold
 - b) fire
 - c) aether
 - d) salt

PHYSICS

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, when scholars began to study light and motion. Alhazen's main work, translated into Latin, was about optics. Alhazen showed that we see because light reflected from objects enters the eye. In the thirteenth century the Silesian scholar Witelo wrote "Perspectiva" – a work on optics. Roger Bacon stressed that certain knowledge comes above all through experience (experiment). Around 1304 Theodoric of Freiberg explained the rainbow by the refraction and reflection of light in raindrops. Musicians knew that if you shorten a vibrating string, its sound will be higher. Builders lifted stones with pulleys, ropes and winches. A compass needle lines up along the (magnetic) north–south line. In the fourteenth century Nicole Oresme wondered whether the daily motion of the sky came from the fact that the Earth rotates on its axis. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 If you shorten a vibrating string, its sound will be...
 - a) higher
 - b) half as loud but the same pitch
 - c) the same
 - d) lower
2. 👁 Theodoric of Freiberg (c. 1304) explained how rainbows form through...
 - a) light reflecting off dark clouds
 - b) dyes floating in the air
 - c) the glow of the full Moon
 - d) refraction and reflection of light in raindrops
3. 📖 What field is the subject of the main work of Alhazen (Ibn al-Haytham), translated into Latin?
 - a) Astronomy
 - b) Optics
 - c) Medicine
 - d) Music
4. 👁 Alhazen showed that we see objects because...
 - a) light reflected from objects enters the eye
 - b) the air carries images
 - c) the soul touches things
 - d) the eyes send out rays
5. 🏰 Medieval builders lifted heavy stones using...
 - a) pulleys, ropes and winches
 - b) balloons
 - c) magnets
 - d) steam
6. 🧑 What did Witelo, a 13th-century scholar from Silesia, write about in his work "Perspectiva"?
 - a) Law
 - b) Optics
 - c) Music
 - d) Agriculture
7. ❤ Roger Bacon (13th century) stressed that certain knowledge is gained above all through...
 - a) experience (experiment)
 - b) dreams
 - c) blind trust in authorities
 - d) drawing lots
8. 🧭 A compass's magnetic needle aligns itself...
 - a) along the (magnetic) north–south line
 - b) toward the nearest sea
 - c) always toward the Sun
 - d) randomly
9. 🌟 Nicole Oresme (14th century) considered whether the daily motion of the sky might be due to the fact that...
 - a) the Sun is fixed at the center
 - b) the Earth is flat
 - c) the stars are attached to the Moon
 - d) the Earth rotates on its axis

ENVIRONMENT

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to a world of great forests and fickle weather. In old Polish, "puszcza" meant a vast, wild forest. The Białowieża Forest was above all a hunting ground of the grand dukes. Hermits chose a life in solitude, far from people. A warmer period around 950–1250 helped the Vikings settle in Greenland. The Vikings also dragged their boats overland between rivers – such a place is called a portage. During storms church bells were rung in the belief that this would drive away the storm and hail. In 1315–1317 the Great Famine was the result of a series of rainy, cold years and crop failures. During the Black Death the plague was spread mainly by fleas, including those living on rats. And in 1979 St. Francis of Assisi was proclaimed the patron of ecologists. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏰 In the Middle Ages, church bells were rung during storms in the belief that...
 - a) this would drive away the storm and hail
 - b) it would summon the knights
 - c) it would calm the animals
 - d) it would attract rain
2. 🌐 A warmer period around 950–1250 made it easier for the Vikings to settle in...
 - a) Madagascar
 - b) Greenland
 - c) Antarctica
 - d) the Sahara
3. 📖 In old Polish, the word "puszcza" meant...
 - a) a vast, wild forest
 - b) a desert
 - c) a riverside meadow
 - d) a monastery garden
4. 🏰 The Great Famine in Europe of 1315–1317 was caused by...
 - a) a plague of locusts in England
 - b) a series of rainy, cold years and crop failures
 - c) the Hundred Years' War
 - d) a volcanic eruption in Poland
5. 🚶 The Vikings dragged their boats overland between rivers. Such a place is called a...
 - a) ford
 - b) causeway
 - c) lock
 - d) portage
6. 🏰 In the Middle Ages, the Białowieża Forest was above all...
 - a) a site of great fairs
 - b) a hunting ground of the grand dukes
 - c) a wasteland without animals
 - d) a cannon testing range
7. 💖 Medieval hermits chose a life...
 - a) on Hanseatic merchant ships
 - b) at the royal court in the capital
 - c) in the most crowded cities of Italy
 - d) in solitude, far from people
8. 🍃 During the Black Death, plague was spread mainly by...
 - a) dogs
 - b) mosquitoes
 - c) flies
 - d) fleas, including those living on rats
9. ⭐ In 1979, St. Francis of Assisi was proclaimed the patron saint of...
 - a) sailors
 - b) ecologists
 - c) doctors
 - d) miners

HEALTH

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages and see how people cared for their health. Hildegard of Bingen wrote about herbs, but she was also a composer of sacred music. Public baths were popular places for bathing and socializing. Fasting meant mainly eating less, especially less meat. People with leprosy were isolated in hospitals called leprosaria. Ergot on rye caused a disease called St. Anthony's fire. The name "bubonic plague" comes from buboes – painful, swollen lymph nodes. In 1377 Ragusa, today's Dubrovnik, ordered arrivals from plague areas to stay isolated before entering the city. The word "quarantine" comes from the Italian "quaranta giorni," meaning forty days. And in the fifteenth century the dance of death reminded everyone that all are equal in death: king, bishop and peasant. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏰 Hildegard of Bingen wrote about herbal healing, but she was also a...
 - a) ship's captain
 - b) composer of sacred music
 - c) clockmaker
 - d) Queen of France
2. 👁 The name "bubonic plague" comes from buboes, which are...
 - a) painful, swollen lymph nodes
 - b) smoke from burning houses
 - c) a rash on the hands
 - d) black spots in the sky
3. 🗨 The word "quarantine" comes from the Italian "quaranta giorni," meaning...
 - a) four weeks
 - b) three months
 - c) forty days
 - d) ten days
4. 🚩 In 1377, Ragusa (Dubrovnik) ordered newcomers from plague-stricken areas to...
 - a) wear red cloaks
 - b) pay double customs duties
 - c) stay isolated before entering the city
 - d) pray at the gate
5. 🛀 Public bathhouses in medieval towns were...
 - a) always built inside castle walls
 - b) open only to kings and princes
 - c) popular places for bathing and socializing
 - d) strictly forbidden by town law
6. 🏥 Hospitals for people with leprosy were called...
 - a) student hostels
 - b) knightly hospices
 - c) leprosaria
 - d) town infirmaries for students
7. 💜 Medieval fasting consisted mainly of...
 - a) sleeping standing up for 40 days
 - b) a total ban on drinking water
 - c) eating only meat and cheese
 - d) eating less, especially less meat
8. 🌿 Ergot, a fungus that infects rye, caused a disease called...
 - a) St. Anthony's fire
 - b) knight's fever
 - c) smallpox
 - d) scurvy
9. 🌟 The "dance of death" (danse macabre) motif in 15th-century art reminded people that...
 - a) all are equal in death: king, bishop and peasant
 - b) everyone dances in paradise after death
 - c) dancing cures disease
 - d) death passes over the rich

1.b.2.a.3.c.4.c.5.c.6.c.7.d.8.a.9.a

A N S W E R S

PHYSICAL CULTURE AND SPORT

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to tournament grounds and hunting trails. The knightly ideal of upbringing combined physical fitness with virtue, courtly manners and piety. The word "tournament" comes from the French "tournoi," linked to "tourner," to turn around. In a joust, a knight scored points when he broke his lance on his opponent. The Truce of God banned fighting on certain days of the week and on feast days. English archers with longbows, as at Crécy in 1346, were famous for their draw strength and rapid rate of fire. In Polish marksmen's guilds, the winner of shooting at a wooden bird became the fowler king. On the hunt, greyhounds chased game mainly by sight. In chess the fers, forerunner of the queen, moved only one square diagonally. People also danced the lively saltarello, which came from Italy. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 Which country does the lively medieval dance saltarello come from?
 - a) England
 - b) Italy
 - c) Poland
 - d) Sweden
2. 👁 In medieval chess, the piece that became today's queen (the fers) moved...
 - a) by jumping like a knight
 - b) only one square diagonally
 - c) only backward
 - d) any number of squares in any direction
3. 📖 The word "tournament" comes from the French "tournoi," linked to the verb "tourner," meaning...
 - a) to race
 - b) to celebrate
 - c) to turn around
 - d) to fight with a sword
4. 🏇 In a tournament joust, points were scored by a knight who, among other things,...
 - a) shouted loudest before charging
 - b) broke his lance on his opponent
 - c) had the finest coat of arms
 - d) dismounted first
5. 🏹 English longbowmen, for example at Crécy in 1346, were famous for their...
 - a) shooting from horseback
 - b) draw strength and rapid rate of fire
 - c) iron bows
 - d) poisoned arrows
6. 🏹 In the fowler brotherhoods of Polish towns, burghers shot at a wooden bird, and the winner became...
 - a) the army commander
 - b) the fowler king
 - c) the town reeve
 - d) a bishop
7. 💖 The knightly ideal of education combined physical fitness with...
 - a) strength alone
 - b) trade
 - c) the study of astronomy
 - d) virtue, courtly manners and piety
8. 🐾 During medieval hunts with greyhounds, the dogs...
 - a) swam after fish
 - b) tracked only by scent underground
 - c) guarded the horses
 - d) chased game mainly by sight
9. ⭐ The Truce of God (Treuga Dei) forbade fighting...
 - a) in port cities and their harbors
 - b) on certain days of the week and on feast days
 - c) during the grain harvest season
 - d) on the birthday of the reigning king

MEDICINE

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to a physician's chamber and a hospital hall. According to humoral theory, illness was the result of an imbalance of the four humors. Around 1025 Avicenna, or Ibn Sina, wrote "The Canon of Medicine." A popular diagnosis was uroscopy – judging the color and appearance of urine. Bloodletting and minor procedures were often done by barber-surgeons. Trotula of Salerno was a physician focused on women's health, and Moses Maimonides, the sultan's doctor, was also a philosopher, author of "The Guide for the Perplexed." The Knights Hospitaller were founded in Jerusalem to care for pilgrims and the sick. Leprosy is caused by leprosy bacilli (bacteria), though they were discovered only in the nineteenth century. Lepers had to warn of their approach with the sound of a clapper or a bell. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 In the Middle Ages, lepers had to warn others of their approach with the sound of...
 - a) a clapper or a bell
 - b) a whistle
 - c) a trumpet
 - d) a drum
2. 👁 Uroscopy, a popular method of diagnosis in the Middle Ages, involved examining...
 - a) the shape of clouds over the town
 - b) the lines on the patient's palms
 - c) the patient's dreams from last night
 - d) the color and appearance of urine
3. 📖 Who wrote "The Canon of Medicine" (c. 1025), a textbook used at European universities for centuries?
 - a) Galen
 - b) Hippocrates
 - c) Paracelsus
 - d) Avicenna (Ibn Sina)
4. ⚖ According to humoral theory, illness was the result of...
 - a) bacterial infection from water
 - b) lightning strikes near the house
 - c) a lack of vitamins in the diet
 - d) an imbalance of the four humors
5. ✂ Bloodletting, shaving and minor procedures were often performed by...
 - a) monastic scribes
 - b) blacksmiths
 - c) barber-surgeons
 - d) heralds
6. 👩 Trota of Salerno (11th–12th centuries) was...
 - a) a queen of Sicily
 - b) a merchant's wife from Venice
 - c) a physician focused on women's health
 - d) a love poet
7. ❤ Moses Maimonides (12th century), physician to the sultan in Egypt, was also an outstanding...
 - a) architect of mosques in Cairo
 - b) sailor and discoverer of islands
 - c) philosopher, author of "The Guide for the Perplexed"
 - d) composer of courtly songs
8. 🌿 Leprosy is caused by...
 - a) leprosy bacilli (bacteria)
 - b) flu viruses
 - c) staying in the sun too long
 - d) waterborne parasites
9. ☆ The Order of St. John (the Hospitallers) was founded in Jerusalem at a hospital to care for...
 - a) pilgrims and the sick
 - b) merchants
 - c) castles
 - d) falcons

PHARMACY

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to monastery gardens and the first pharmacies. The life of monks growing herbs followed the rhythm of the canonical hours, the prayers. In the ninth century Abbot Walafrid Strabo described his garden in the poem "Hortulus." Around 800 Charlemagne, in the "Capitulare de villis," ordered his estates to grow a long list of plants, including herbs. The word "apothecary" comes from the Greek "apotheke," a storehouse. In a mortar the apothecary ground and mixed the ingredients of medicines. Theriac, a remedy of dozens of ingredients, was meant to be a universal antidote to poisons and illnesses. The Edict of Salerno of Emperor Frederick II separated the apothecary's profession from the physician. The symbol of pharmacy is the Bowl of Hygieia with a snake, and the patrons of pharmacists are the twins Cosmas and Damian. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Life in a monastery, where monks tended herb gardens, followed the rhythm of...**
 - a) military drums
 - b) harvests and fairs
 - c) the canonical hours (prayers)
 - d) shop bells
2. 👁 **The symbol of pharmacy, a bowl with a snake coiled around it, is the...**
 - a) Bowl of Hygieia
 - b) Olympic cup
 - c) Grail chalice
 - d) horn of plenty
3. 📖 **The word "apothecary" comes from the Greek "apotheke," meaning...**
 - a) poison
 - b) storehouse
 - c) doctor
 - d) herb
4. ⚖ **Theriac, a medieval medicine made of dozens of ingredients, was supposed to work as...**
 - a) a universal antidote to poisons and illnesses
 - b) lamp fuel
 - c) a paint for books
 - d) a sleeping draught for horses
5. ✂ **In a mortar, the apothecary...**
 - a) measured the hours of the day
 - b) ground and mixed the ingredients of medicines
 - c) weighed out gold for customers
 - d) cooked soup for the sick
6. 👤 **The Edict of Salerno by Emperor Frederick II (1231/1240) separated the profession of apothecary from that of...**
 - a) physician
 - b) merchant
 - c) blacksmith
 - d) priest
7. 💖 **Walafrid Strabo, abbot of Reichenau (9th century), described his monastery herb garden in...**
 - a) a map of the world
 - b) a letter to the pope
 - c) a war chronicle
 - d) the poem "Hortulus"
8. 🌿 **In the "Capitulare de villis" (c. 800), Charlemagne ordered his estates to grow...**
 - a) only roses for the court gardens
 - b) only grapevines for wine
 - c) only wheat for bread
 - d) a long list of plants, including herbs
9. ⭐ **Which twin saints, doctors who healed free of charge, are the patrons of pharmacists?**
 - a) Cyril and Methodius
 - b) Gervasius and Protasius
 - c) Peter and Paul
 - d) Cosmas and Damian

AGRICULTURE

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to the village fields. In the "Très Riches Heures du Duc de Berry," the calendar miniatures show farm work in each month of the year. The heavy moldboard plow let farmers turn over the sod in heavy soils. In the three-field system one field was sown with winter grain, another with spring grain, and the third lay fallow so the soil could rest. In Poland a "łan" was a unit of land area. Settlers under German law received a period of several years' exemption from rent. After harvest peasants paid the tithe – a tenth of the crops given to the Church – and the end of harvest was celebrated with Dożynki and a wreath of grain. In the late Middle Ages buckwheat arrived from Asia. The patron of farmers is St. Isidore, the plowman of Madrid. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **What is the traditional Polish end-of-harvest festival with songs and a wreath of grain ears called?**
 - a) Zaduszki
 - b) Andrzejki
 - c) Topienie Marzanny
 - d) Dożynki
2. 👁 **The calendar miniatures in the "Très Riches Heures du Duc de Berry" (15th century) show...**
 - a) portraits of the Avignon popes
 - b) star maps of the sky
 - c) farm work in each month of the year
 - d) naval battles with the English
3. 🗨 **In medieval Poland, the word "tan" meant...**
 - a) a type of heavy plow
 - b) the yield of one field
 - c) a damp riverside meadow
 - d) a unit of land area
4. 🔄 **In three-field rotation, one field was sown with winter grain, a second with spring grain, and the third...**
 - a) became a permanent pasture
 - b) was flooded with water
 - c) lay fallow so the soil could rest
 - d) was sown with salt
5. 🛖 **The heavy moldboard plow, which spread in the Middle Ages, made it possible to...**
 - a) sow seeds from the air
 - b) harvest grain without a sickle
 - c) dig wells
 - d) turn over the sod in heavy soils
6. 🧑 **Settlers founding a village under German law usually received a "wolnizna," which was...**
 - a) a seat on the town council
 - b) several years' exemption from rent
 - c) a free horse
 - d) the right to hunt aurochs
7. 💖 **The tithe that peasants paid after the harvest was...**
 - a) a tenth of the crops given to the Church
 - b) ten days of work for the king
 - c) every tenth horse in the stable
 - d) ten coins for the reeve
8. 🌾 **Buckwheat reached Europe in the late Middle Ages from...**
 - a) Antarctica
 - b) Asia
 - c) America
 - d) Australia
9. ⭐ **Which saintly plowman from Madrid (12th century) is the patron of farmers?**
 - a) St. Florian
 - b) St. Isidore
 - c) St. Nicholas
 - d) St. Hubert

VETERINARY SCIENCE

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages, to stables, forges and pastures. Shepherd's called their flocks by playing shepherd's horns and trumpets. The word "veterinary" comes from the Latin "veterinae," meaning draft and pack animals. The horseshoe, which spread at that time, protected horses' hooves from wear. It was essential in damp Northern Europe, because moisture softens hooves, which then wear faster. Shoeing means nailing a horseshoe to the hoof. The patron of blacksmiths and horses is St. Eligius, a goldsmith by trade. In the thirteenth century Jordanus Ruffus, marshal of Frederick II's court, wrote a handbook on treating horses. Rabies spreads mainly through the bite of an infected animal. And according to "The Little Flowers of St. Francis," the saint tamed the fierce wolf of Gubbio. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Shepherds called their flocks and signaled to each other in the pastures by playing...**
 - a) lutes
 - b) church bells
 - c) shepherds' horns and trumpets
 - d) organs
2. 👁 **The horseshoe, which spread in medieval Europe, protected...**
 - a) a knight's legs
 - b) doors from thieves
 - c) horses' hooves from wear
 - d) cart wheels
3. 🗨 **The word "veterinary" comes from the Latin "veterinae," meaning...**
 - a) draft and pack animals
 - b) stables
 - c) old doctors
 - d) diseases
4. ➡ **Horseshoes became essential especially in damp Northern Europe because...**
 - a) roads were made of ice
 - b) horseshoes brought luck
 - c) moisture softens hooves, which then wear faster
 - d) horses there had no hooves
5. 🛠 **Shoeing a horse means...**
 - a) gluing a leather sole to the hoof
 - b) nailing a horseshoe to the hoof
 - c) strapping on wooden blocks
 - d) wrapping its legs in linen strips
6. 🧑 **Jordanus Ruffus, marshal of Emperor Frederick II's court, wrote a 13th-century manual on treating...**
 - a) sheep
 - b) hunting dogs
 - c) falcons
 - d) horses
7. ❤ **According to "The Little Flowers of St. Francis," the saint...**
 - a) drove all snakes out of Italy
 - b) tamed the fierce wolf of Gubbio
 - c) taught a donkey to talk
 - d) trained a lion to fight
8. 🍃 **Rabies spreads to people and animals mainly through...**
 - a) mosquito bites
 - b) smell
 - c) the bite of an infected animal
 - d) drinking milk
9. ☆ **Which 7th-century saint, a goldsmith by trade, is the patron of blacksmiths and horses?**
 - a) St. Lawrence
 - b) St. Eligius
 - c) St. Martin
 - d) St. George



ANIMAL SCIENCE AND FISHERIES

5. Middle Ages

500–1500 AD

Let's travel back to the Middle Ages, to ponds, seas and falconry hunts. Small bells tied to a falcon's legs let the falconer hear where the bird was. Fish ponds were made by holding back water with an earthen dike with a sluice. Ponds at Cistercian monasteries were used mainly for raising carp and other fish, because during Lent people ate fish instead of meat. The symbol of the first Christians was a fish – the Greek "ichthys" is an acronym for Jesus Christ, Son of God, Savior. The patron of fishermen, St. Peter, was a fisherman before he became an apostle. The great autumn herring catches took place off Scania in southern Sweden, and Hanseatic merchants preserved the herring by salting it in barrels. England's most valuable export, sold to Flanders, was sheep's wool. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏹 **Small bells tied to falcons' legs let the falconer...**
 - a) count its wingbeats
 - b) hear where the bird was
 - c) calm the bird with music
 - d) scare away other birds
2. 🐟 **Medieval fishponds were created by damming water with...**
 - a) ice
 - b) a brick wall around a lake
 - c) a concrete dam
 - d) an earthen dike with a sluice
3. 🗨 **The Greek word "ichthys" (fish), a symbol of the first Christians, is an acronym for the words...**
 - a) Jesus Christ, Son of God, Savior
 - b) Bread and wine
 - c) Peace be with you
 - d) The Lord is my shepherd
4. 🐟 **Ponds next to monasteries, for example Cistercian ones, were mainly used for raising...**
 - a) sea trout
 - b) crayfish for export to China
 - c) carp and other fish
 - d) whales
5. 🐟 **Herring caught by Hanseatic merchants was preserved for transport by...**
 - a) packing it in ice from refrigerators
 - b) storing it in honey
 - c) drying it in the tropical sun
 - d) salting it in barrels
6. 🐟 **The great autumn herring catches, key to Hanseatic trade, took place off the coast of...**
 - a) Iceland
 - b) Scania (southern Sweden)
 - c) Portugal
 - d) Crimea
7. ❤ **During Lent, instead of meat, medieval Christians ate...**
 - a) poultry
 - b) game
 - c) fish
 - d) beef
8. 🌿 **The most valuable export of medieval England, sold to Flanders, was...**
 - a) sheep's wool
 - b) camel hide
 - c) cotton
 - d) silk
9. ⭐ **Before he became an apostle, St. Peter, patron of fishermen, was a...**
 - a) tax collector
 - b) fisherman
 - c) soldier
 - d) carpenter

OTHER (NATURE)

5. Middle Ages

500–1500 AD



Let's travel back to the Middle Ages and see nature through the eyes of its scholars. The idea of the "book of nature" held that nature is like a book through which one can know the Creator. The word "calendar" comes from the Latin "calendae," the first day of the month. In the Julian calendar a leap year came every 4 years. Easter is celebrated on the Sunday after the first full moon of spring. The English round "Sumer is icumen in" welcomes summer with the song of the cuckoo. The Hereford Mappa Mundi was painted on a single sheet of parchment. The Vikings navigated by watching the Sun, stars and seabirds. Peas were on the table, while potatoes arrived from America only later. And at the very end of the era, Vasco da Gama's expedition became the first to sail from Europe around Africa to India. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The medieval English round "Sumer is icumen in" (13th century) welcomes summer with the song of the...
 - a) nightingale
 - b) lark
 - c) rooster
 - d) cuckoo
2. 🗺️ The Hereford Mappa Mundi (c. 1300) is a world map painted on...
 - a) a church wall
 - b) a wooden panel
 - c) a single sheet of parchment
 - d) a stone slab
3. 📅 The word "calendar" comes from the Latin "calendae," meaning...
 - a) a leap year
 - b) the Christmas holidays
 - c) the new moon in March
 - d) the first day of the month
4. 📅 How often did a leap year occur in the Julian calendar (every how many years)?

5. 🚢 On the open sea, the Vikings navigated by observing, among other things,...
 - a) GPS compass readings
 - b) lighthouses in America
 - c) the Sun, stars and seabirds
 - d) satellites orbiting the Earth
6. 🌐 Vasco da Gama's expedition (1497–1498) was the first to sail from Europe around Africa to...
 - a) China
 - b) Brazil
 - c) India
 - d) Australia
7. 💚 The medieval idea of the "book of nature" held that...
 - a) nature has no meaning
 - b) books grow on trees
 - c) nature is like a book through which one can know the Creator
 - d) all animals must be written down
8. 🌿 Which of these plants was known in medieval Europe before 1492?
 - a) Tomatoes
 - b) Corn
 - c) Peas
 - d) Potatoes
9. ⭐ According to a rule set already in antiquity, Easter is celebrated...
 - a) on the first Sunday of April
 - b) on the day of the solstice
 - c) on the Sunday after the first full moon of spring
 - d) always on March 25